



Noise analysis and efficiency improvement of a pulse-width modulated permanent magnet synchronous motor by dynamic error budgeting[☆]

Thomas Riel^{*}, Rudolf Saathof, Andelko Katalenic, Shingo Ito, Georg Schitter

Automation and Control Institute (ACIN), TU Wien, Gußhausstraße 27-29, Vienna, 1040, Austria

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ABSTRACT

This paper presents the mechatronic analysis and performance estimation of a pulse-width modulated permanent magnet synchronous motor (PMSM), which is part of a direct-drive rotational servo system. Physical modeling together with dynamic error budgeting are applied to identify the minimal required switching frequency that maximizes the energy efficiency of the system while maintaining the required precision. The analysis is carried out for a wide velocity range and considers various friction phenomena. The disturbances induced by the switched amplifier are also considered and compared to a linear amplifier. The accuracy of the derived model and analysis is experimentally verified over the entire velocity range as well as for the considered range of switching frequencies.

1. Introduction

Direct-drive PMSMs are widely used in precision positioning applications, such as robots [1] or optical measurement platforms [2]. Their high accuracy is achieved by eliminating or reducing the negative effects of gear boxes such as backlash and friction. They also have the advantage of high power density and high dynamic performance due to decreased inertia [3], also making them good candidates for electric vehicle [4]. However, to achieve a precise motion with errors in the sub- μ rad range, a detailed analysis of present noise sources and a good mechatronic design of the system are necessary [5].

One of the most precise ways of controlling a PMSM is with vector based control, such as field-oriented control (FOC) [6]. The analysis and compensation of systematic error sources, such as current measurement errors or torque ripples, have attracted quite some attention [7–9]. For precision positioning applications it is important to understand in detail and to minimize all of these systematic disturbances. However, the influence of stochastic disturbances acting in the drive system also needs to be investigated. If not considered in the system design, these disturbances fundamentally limit the achievable precision.

Recently, the analysis of stochastic disturbances acting on the FOC system of a PMSM has been presented [10,11]. The analysis is based on dynamic error budgeting (DEB), using power spectral densities (PSDs) of the known noise sources. However, the analysis is performed for a linear power amplifier, considering only two different angular velocities.

Linear power amplifiers are regularly used in precision positioning applications due to their low noise. However, they suffer from high energy dissipation and therefore have low efficiency. This is problematic, especially in space constrained or highly integrated applications, as the additional heat intake requires a larger heat sink and leads to unwanted thermal effects. Mitigating these thermal issues requires additional cooling effort, which may induce additional disturbances, such as vibrations caused by the cooling system. Switched power amplifiers relax these thermal issues due to their high efficiency. In contrast to linear amplifiers, which show a typical efficiency below 50%, switched amplifiers reach efficiencies of more than 90% [12]. Switched amplifiers using (pulse-width modulation) are commonly employed in current controlled power amplifiers for PMSM. However, they suffer from higher noise, which may degrade the system accuracy in precision positioning applications. To design an energy efficient precision servo system it is therefore important to identify, analyze, and reduce the disturbances induced by switched amplifiers.

Switched amplifiers also show substantial noise at frequencies below the fundamental switching frequency [12,13]. As most mechatronic systems are more susceptible to noise below practical PWM switching frequencies, special emphasis has to be put on this frequency region. The primary sources of this low frequency noise are power supply noise [14], inter-modulation distortions [15], and sampling and quantization effects [12,16,17]. Considering these effects in the analysis of a drive system with switched amplifier is necessary to provide an accurate description. A simplified analysis of the induced

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^{*} Corresponding author.

E-mail address: riel@acin.tuwien.ac.at (T. Riel).

disturbances is possible by only considering the fundamental switching frequency of the PWM [18]. However, this kind of analysis does not include the entire PWM signal spectrum, which includes a variety of spectral components [19], and neglects the adverse low frequency effects.

This contribution therefore aims to analyze all relevant, stochastic disturbances acting in the FOC system of a PMSM with switched power amplifiers. In addition, the analysis is performed for a continuous velocity range, including friction effects and torque distortions of the ball bearings. The analysis is used to investigate the tradeoff between the system precision and amplifier efficiency, in order to identify the ideal switching frequency. The analysis is performed using DEB for a continuous operation range from standstill to several degrees per second and includes important friction phenomena, which is required for low velocity, high precision positioning [11]. To our best knowledge this is the first time such an analysis of a switched system using DEB is presented.

The remainder of this paper is organized as follows. Section 2 starts by reviewing the theoretical foundation and necessary assumptions for DEB. Section 3 gives a short description of the analyzed system, which is mathematically modeled in Section 4. The derived models are utilized in Section 5 to analyze the influences of the stochastic disturbances by DEB. In Section 6 the performance analysis with corresponding experimental validations is presented. Finally, the conclusion is given in Section 7.

2. Review of the theoretical foundation of dynamic error budgeting

DEB is a method for estimating the performance of a mechatronic system, by modeling the stochastic disturbances acting on the system by their corresponding PSD [20]. DEB has been applied to disk drives [21], magnetically suspended rotating platform [22] and rotational servo systems [10,11]. Before applying DEB, it is important to know its necessary assumptions and limitations. Therefore, the theoretical foundation of DEB is reviewed in this section. Based on this review, the necessary assumptions for the application of DEB are derived.

2.1. Linking the time domain to the frequency domain

In contrast to previous work [20,22], this paper uses the Wiener–Khinchin theorem to link the time domain and the frequency domain, instead of Parseval’s theorem [23]. This is necessary because the Fourier transform of a random variable does not exist as a random variable, in general, has infinite energy, and therefore Parseval’s theorem can only be applied to a truncated version of a stochastic process [23]. In engineering practice, the use of a truncated version of a stochastic process may be the preferred choice, however, for modeling purposes a precise definition may be required.

Assume $x(t)$ is a complex, wide-sense stationary, random process, then the PSD of the random process, $PSD_x(\omega)$, is defined by [24]

$$R_x(\tau) = \int_{-\infty}^{\infty} e^{i\omega\tau} PSD_x(\omega) d\omega, \quad (1)$$

where $R_x(\tau)$ is the auto-correlation function of the random process. Hereby, Eq. (1) is known as the Wiener–Khinchin relation [23].

If the inverse Fourier transform of $R_x(\tau)$ exists, the PSD can directly be calculated from [24]

$$PSD_x(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega\tau} R_x(\tau) d\tau. \quad (2)$$

This means the auto-correlation function $R_x(\tau)$ and the PSD of the random process $PSD_x(\omega)$ form a Fourier pair.

If it is further assumed that $x(t)$ is an *ergodic process*, then $R_x(\tau)$ can be calculated from measurements in the time domain, by forming the long-time average of the auto-correlation function [24]

$$R_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t+\tau)dt. \quad (3)$$

Eqs. (2) and (3) and the assumption of an ergodic, wide-sense stationary, random process are therefore sufficient to link the time domain and the frequency domain for the application of DEB.

For practical issues it may be favorable to redefine the random process $x(t)$ as sum of its random varying part $x_1(t)$ and its mean value μ [24]

$$x(t) = \mu + x_1(t), \quad (4)$$

thereby avoiding that the PSD has spectral content at $\omega = 0$.

2.2. Power spectral density conversion of linear systems

For linear-time invariant (LTI) systems the PSD of the output y of the system $PSD_{y,x}(s)$, induced by a disturbance x at its input is given by [23]

$$PSD_{y,x}(s) = |T_{y,x}(s)|^2 PSD_x(s), \quad (5)$$

where $PSD_x(s)$ is the PSD of the input and $T_{y,x}(s)$ is the transfer function from x to the output y .

By assuming independence between the different disturbances acting on a system, the overall PSD of the output y can be found by summation of the effect of all disturbances [23]

$$PSD_y(s) = \sum_i |T_{y,x_i}(s)|^2 PSD_{x_i}(s). \quad (6)$$

If the independence is not given, then the covariance between the different disturbances has to be taken into account [23]

$$PSD_y(s) = \sum_i |T_{y,x_i}(s)|^2 PSD_{x_i}(s) + \sum_{i \neq j} |T_{y,x_i}(s)| |T_{y,x_j}(s)| CSD_{x_i,x_j}(s), \quad (7)$$

where $CSD_{x_i,x_j}(s)$ is the cross-spectral density (CSD) between the two inputs x_i and x_j .

3. System description

The system to be investigated consists of two individual rotational axis with direct-drive PMSM, as shown in Fig. 1. The custom build PMSMs have eleven pole pairs and a rated torque of 6.44 Nm. The servo system is used to position an optical imaging system with high precision. Two ball bearings are used to accommodate the load of the imaging system. To minimize the influence of the positioning error on the optical measurement result, a low positioning error is required. For

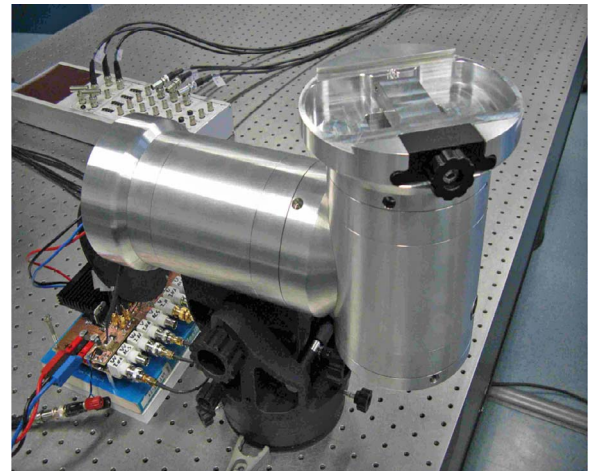


Fig. 1. Laboratory prototype of the two axis rotational servo system with direct-drive PMSMs.

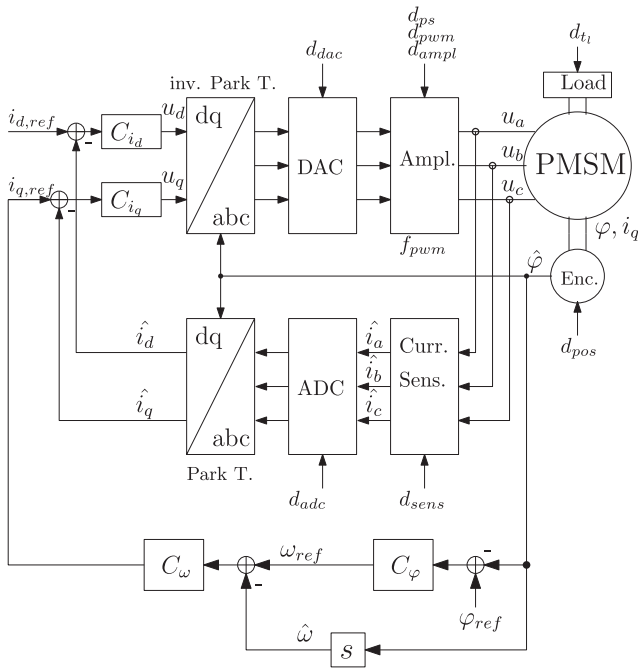


Fig. 2. Schematic representation of the FOC system of a PMSM with the individual components and the considered disturbance (denoted by d_x) sources acting on them.

example, high performance optical tracking systems aim for a root mean square (RMS) position error that is below the atmospheric seeing conditions [25]. Under excellent climate conditions, the atmospheric seeing ranges from 0.4 to 0.7'' [26]. Targeting a factor of five better performance, this corresponds to a required precision of 680 nrad. This level of precision has to be maintained for a motion with a velocity of up to one degree per second. This is equivalent to the maximum angular velocity required for observing the international space station (ISS) when passing up to 10° below ones zenith. Similar requirements are targeted in other applications of such servo systems.

In the targeted application, the servo system performs a continuous motion at quasi constant velocity. In addition to the targeted velocity range from 0.01 to 1°/s, a rapid repositioning with slew velocities of up to 20°/s is also desired. This wide required velocity range makes it necessary to include friction phenomena into the physical model.

4. System modeling

The considered FOC system of a PMSM with the individual components and the considered disturbance sources is shown in Fig. 2. The system consists of a power amplifier (Ampl.), which supplies the three phase voltages (u_a, u_b, u_c) to the PMSM. The resulting phase currents (i_a, i_b, i_c) are measured by current sensors and discretized by three 16-Bit, 1 Msps ADS channels. The PMSM is directly connected to the load, which is subject to torque disturbances. The angular position of the rotor φ is measured by an optical encoder (Enc.), which is sampled at a rate of 20 kHz. The current control system and PWM modulation is implemented on the FPGA of a dSpace Rapid Prototyping System (DS1202, dSpace GmbH, Germany). A cascaded velocity and position control system is implemented on the processor of the same prototyping system.

4.1. Physical model

To analyze the performance of the system, an accurate model of the rotational servo axis is required. The system is split into an electrical and mechanical subsystem for modeling [11].

The mechanical system is described as a lumped mass model

$$\frac{d\varphi}{dt} = \omega \quad (8)$$

$$\frac{d\omega}{dt} = \frac{1}{J}(T_e - T_f - T_l), \quad (9)$$

where φ is the mechanical rotor angular position, ω the mechanical rotor angular velocity, J the moment of inertia, T_e the electrical torque generated by the PMSM, T_f the torque due to the friction force F_f , created by the used ball bearings, and T_l is a variable external load torque, which is considered as additional input in the further analysis.

For the investigated velocity range, friction is one of the main mechanical disturbance source. The LuGre-Model [27] is used to model the friction behavior. Compared to alternative friction-models [28–30], this model has the advantage of having a small number of parameters with well-defined physical meaning. The parameters of the LuGre-Model also easily translate to the parameters of common static friction models when linearized around a given velocity. The complete friction model is given by

$$\frac{dz}{dt} = \omega - c_0 \frac{|\omega|}{g(\omega)} z \quad (10)$$

$$F_f = c_0 z + c_1 \dot{z} + c_2 \omega, \quad (11)$$

with

$$g(\omega) = F_c + (F_s - F_c)e^{-(\omega/\omega_s)^2}, \quad (12)$$

where z is the average bristle deflection, ω the velocity between the sliding surfaces, F_f the resulting friction force, c_0 represents the stiffness of the bristles, c_1 the microdamping of the bristles, c_2 the macroscopic viscous friction, F_s and F_c represent the stiction and the Coulomb friction force, and ω_s is a parameter to model the Stribeck effect.

The mechanical motion of an individual axis is therefore modeled by Eqs. (8)–(11).

To describe the complete electromechanical system, an electrical model of the PMSM is used [11]. To simplify the electrical system, all quantities in the stator coordinate system are transformed into the rotor coordinate system by means of the power invariant Park transformation [10].

The resulting electrical system in rotor coordinates is described by

$$\frac{di_d}{dt} = \frac{1}{L_d}(u_d - R i_d + \omega_e L_q i_q) \quad (13)$$

$$\frac{di_q}{dt} = \frac{1}{L_q}(u_q - R i_q + \omega_e L_d i_d - \omega_e \Psi_M). \quad (14)$$

The electrical currents result in a corresponding electric torque given by

$$T_e = \frac{3}{2} Z_p (\Psi_M i_q + (L_d - L_q) i_d i_q), \quad (15)$$

where L_d, L_q are the d, q axis inductances, u_d, u_q the d, q axis stator voltages, R the stator resistance, i_d, i_q the d, q axis stator currents, Ψ_M the mutual magnetic flux due to the rotor magnets, Z_p the number of pole pairs and $\omega_e = Z_p \omega$ the electrical velocity.

The complete electromechanical system is modeled by Eqs. (8)–(15).

To use this non-linear model for the performance analysis within the DEB framework, as discussed in Section 2, it has to be linearized around a given velocity. With $\omega = \omega_0 = \text{constant}$ and the state, $\mathbf{x} = [\omega, i_d, i_q, z]^T$, the system is modeled by

$$\dot{\mathbf{x}} = \mathbf{A} \mathbf{x} + \mathbf{B} \mathbf{u}, \quad (16)$$

where $\mathbf{A}$ is the system matrix, $\mathbf{u} = [u_d, u_q, t_l]^T$ is the input, and $\mathbf{B}$ is the input matrix. The system is described by

$$\mathbf{A} = \begin{bmatrix} -\frac{c_1(1 + \text{sign}(\omega_0)) - c_2}{J} & \frac{(L_d - L_q)(c_2 \omega_0 + F_c)}{J\Psi_M} & \frac{3}{2} \frac{\Psi_M Z_p}{J} & -\frac{c_0}{J} \left(1 - \frac{c_1 |\omega_0|}{F_c}\right) \\ \frac{2}{3} \frac{L_q(c_2 \omega_0 + F_c)}{L_d \Psi_M} & -\frac{R_q}{L_d} & \frac{L_q Z_p \omega_0}{L_d} & 0 \\ -\frac{\Psi_M Z_p}{L_q} & \frac{L_d Z_p \omega_0}{L_q} & -\frac{R_q}{L_q} & 0 \\ 1 - \text{sign}(\omega_0) & 0 & 0 & -\frac{c_0 |\omega_0|}{F_c} \end{bmatrix}, \quad (17)$$

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & \frac{1}{J} \\ \frac{1}{L_d} & 0 & 0 \\ 0 & \frac{1}{L_q} & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (18)$$

Eqs. (17) and (18) represent the linear model of a single rotational axis for a given, constant velocity ω_0 .

4.2. Component models

Each of the considered components in the FOC control system, as shown in Fig. 2, is modeled by a LTI transfer function. Each disturbance source is described by its PSD. If applicable, all disturbances are referred to the output of the component. The model parameters are provided in Table 1.

4.2.1. Amplifier model

The custom made power amplifiers use OPA549 operational amplifiers (Texas Instruments Incorporated, Dallas, USA), and are modeled by a first order transfer function $G_{\text{ampl}}(s)$ with a cut-off frequency $f_{c,\text{ampl}}$ of 600 kHz. The amplifiers are connected in a non-inverting configuration and can be used in linear as well as switched operational mode. In the second case, the amplifier represents a single-feedback switched amplifier, which has the advantage of improved power supply rejection ratio (PSRR) compared to open loop switched amplifiers [13]. The noise induced by the amplifiers is modeled as a combination of 1/f-noise and white noise. In accordance with the datasheet, the 1/f-noise is specified by the noise corner frequency $f_{nc,\text{ampl}}$ of 20 Hz and the white, gaussian noise is specified by the calculated noise power spectral density n_{ampl} .

Table 1

System parameters of the control system used for simulation.

Bandwidth ADC $f_{c,\text{ADC}}$	10 kHz
Bandwidth DAC $f_{c,\text{DAC}}$	110 kHz
Nyquist frequency f_N	500 kHz
LSB voltage q	610 μV
Full scale range FSR	20 V
SNR of ADC SNR_{adc}	88 dB
SNR of DAC SNR_{dac}	80 dB
Bandwidth amplifier $f_{c,\text{ampl}}$	600 kHz
Noise corner amplifier $f_{nc,\text{ampl}}$	20 Hz
Amplifier noise PSD n_{ampl}	107 nV^2/Hz
Amplifier gain A_{ampl}	1.5
PWM Amplitude A_{pwm}	30 V
Power supply noise PSD n_{ps}	500 $\mu\text{V}^2/\text{Hz}$
Bandwidth current sensor $f_{c,\text{sens}}$	10 kHz
Noise corner sensor $f_{nc,\text{sens}}$	10 Hz
Current sensor noise PSD n_{sens}	$168 \times 10^{-12} \text{A}^2/\text{Hz}$
Current sensor gain A_{isens}	1 V/A
Bandwidth encoder $f_{c,\text{pos}}$	5 kHz
Encoder noise PSD n_{pos}	2.93 mrad^2/Hz
Current control bandwidth BW_i	600 Hz
Velocity control bandwidth BW_ω	70 Hz
Position control bandwidth BW_ϕ	10 Hz
Torque fluctuation constant K_N	$2 \times 10^{-15} \text{Nm}/\sqrt{\text{s/rad}}$

$$\text{PSD}_{\text{ampl}}(s) = |G_{\text{ampl}}(s)|^2 n_{\text{ampl}} \left(1 + \frac{(2\pi f_{nc,\text{ampl}})^2}{s^2}\right). \quad (19)$$

In switched operational mode, the modulation of the output voltage forms an additional disturbance to the system. Natural sampling PWM is used in the modulation process, as this form of PWM contains no harmonic distortions in the baseband [19]. The PSD of the modulated signal consists of the sum of three distinct components: A carrier dependent spectrum $P_c(f)$

$$P_c(f) = \sum_{k=0}^{\infty} \frac{2A_{\text{pwm}}}{j\pi(2k+1)} [\delta(f - (2k+1)f_c) - \delta(f + (2k+1)f_c)], \quad (20)$$

a signal dependent spectrum $P_s(f, x)$

$$P_s(f, x) = \sum_{k=1}^{\infty} (-1)^k \sum_{n=1}^{\infty} \frac{(j\pi k)^{n-1}}{n!} [S_n(f + kf_c) + (-1)^{n-1} S_n(f - kf_c)], \quad (21)$$

and the signal spectrum $X(f)$ itself [19]. Hereby, f_c is the PWM frequency, $\delta(\cdot)$ is the Dirac delta function, and $S_n(f)$ are the spectra of the n^{th} power of the input signal $x(t)$. The resulting output spectra is

$$P_{\text{pwm}}(f) = X(f) + P_c(f) + P_s(f, x), \quad (22)$$

where only the first term represents the desired signal, while the other components can be regarded as disturbances to the system. As the PWM represents a non-linear function, the effect of the high frequency disturbances is only approximated. For the approximation it is assumed that the high frequency components are sufficiently filtered by the low pass characteristic of the PMSM and the various components in the current control loop. It can therefore be assumed that the maximal frequency f_s of $X(f)$ is much smaller than the PWM frequency f_{pwm} . Therefore, no inter-modulation occurs, and the PWM can be treated as a constant gain for the feedback control. For the performance analysis, the effect of the high frequency components is included by calculating the spectrum of $X(f)$ and then applying Eq. (21) to get the output spectrum. Hereby only the components of $k = 1$ and $n = 1..5$ are considered in Eq. (21). The high frequency components are then treated as additional disturbance to the linearized system

$$\text{PSD}_{\text{pwm}}(s) = |G_{\text{ampl}}(s)|^2 (P_c(s)^2 + P_s(s, x)^2). \quad (23)$$

Another major disturbance in switched operation mode is the power supply noise [14], which is characterized by the PSRR of the amplifier. As the amplifier is used with voltage feedback, the PSRR is improved in comparison to open loop switched amplifiers, where the PSRR can reach as low as 6 dB [14]. Based on measurements (data not shown), the PSRR was modeled and reaches from 100 dB for frequencies below 1 Hz to 40 dB for frequencies above 1500 Hz.

In addition to the power supply noise that is directly transmitted to the output of the amplifier, also inter-modulation effects occur between the PWM and the power supply noise [15]. This phenomenon is included by using Eq. (21), as described above, with the input signal spectrum $X(f)$ replaced by the spectrum of the power supply noise, resulting in

$$\text{PSD}_{\text{ps,sw}}(s) = |\text{PSRR}(s)|^2 (n_{\text{ps}}^2 + P_s(s, \sqrt{n_{\text{ps}}})^2), \quad (24)$$

where n_{ps} describes the measured noise power spectral density of the power supply.

4.2.2. Current sensor model

The current sensors utilize AMC1301 isolation amplifiers (Texas Instruments Incorporated, Dallas, USA) and shunt resistors to measure the phase currents. The use of an isolation amplifier is necessary to achieve a good common mode rejection ratio (CMRR) against the high PWM voltages. The current sensors are modeled by a first order transfer function $G_{\text{sens}}(s)$ with a cut-off frequency $f_{c,\text{sens}}$ of 10 kHz. To include

low frequency disturbances, the noise of the current sensors is modeled as a combination of $1/f$ -noise and white noise. The noise corner frequency $f_{nc,sens}$ and the noise power spectral density n_{sens} are determined by measurements. The resulting PSD is given by

$$PSD_{sens}(s) = |G_{sens}(s)|^2 n_{sens} \left(1 + \frac{(2\pi f_{nc,sens})^2}{s^2} \right). \quad (25)$$

In switched operational mode of the power amplifier, the high frequency components of the PWM are transmitted by the current sensor. This effect is on the one hand due to the still limited CMRR, especially at high frequencies, and on the other hand due to cross-coupling between the individual signal traces on the PCB. Based on measurements, this effect is modeled by a constant gain $G_{emi}(s)$ of -90 dB between the phase voltages and the measured current signal, as shown in Fig. 2.

4.2.3. Analog-to-digital converter model

The transfer function of the 16-Bit, 1Mps ADC $G_{adc}(s)$ is given by their first order anti-aliasing filter, with a cut-off frequency $f_{c,ADC}$ of 10 kHz. The signal to noise ratio (SNR) together with quantization noise is used to describe the PSD of the ADC. Both noise sources are assumed to follow a white frequency distribution up to the Nyquist frequency $f_N = 500$ kHz. The PSD is given by [20]

$$n_{adc} = \frac{q^2}{12} + \left(\frac{FSR}{2\sqrt{2}} 10^{-SNR_{adc}/20} \right)^2 \quad (26)$$

$$PSD_{adc}(s) = \frac{n_{adc}}{f_N}, \quad (27)$$

where q is the voltage of the (least significant bit), FSR is the (full scale range) and SNR_{adc} is the SNR of the ADC.

4.2.4. Digital-to-analog converter model

The transfer function $G_{dac}(s)$ of the 16-Bit, 1 Mps (DACs) is modeled by their first order reconstruction filter with a cut-off frequency $f_{c,DAC}$ of 110 kHz. The SNR together with quantization noise is used to describe the PSD of the DAC, according to Eqs. (26) and (27).

In switched operation mode, the quantization of the output signal is performed prior to the DAC with a resolution of 10-Bit. The DAC is then used to generate the PWM signal by switching the output between zero and FSR. In switched operation mode, the additional quantization noise is included in the PSD of the DAC by adjusting q . In linear operation mode, the DAC is used to output the reference signal with full resolution.

4.2.5. Encoder model

The transfer function of the encoder $G_{pos}(s)$ is approximated by a first order low-pass filter with a cut-off frequency $f_{c,pos}$ of 5 kHz, and white, gaussian noise, specified by its noise power spectral density n_{pos} of $2.93 \text{ nrad}^2/\text{Hz}$ is assumed for the PSD

$$PSD_{pos}(s) = |G_{pos}(s)|^2 n_{pos}. \quad (28)$$

4.2.6. Current controller model

As it is common practice, PI-controllers are used for controlling the current in the direct as well as the quadrature axis [31]. The corresponding transfer function of the controllers is given by

$$C_{i,q,d}(s) = K_{p,i} + \frac{K_{i,i}}{s}. \quad (29)$$

The gains for both axis are identical and tuned for a closed loop bandwidth of 600 Hz.

4.2.7. Motion controller model

The motion control of the servo system consists of two cascaded control loops: an inner velocity control loop and an outer position

control loop. The corresponding transfer function of the used PI-controllers are given by

$$C_{\omega}(s) = K_{p,\omega} + \frac{K_{i,\omega}}{s} \quad (30)$$

$$C_{\varphi}(s) = K_{p,\varphi} + \frac{K_{i,\varphi}}{s}. \quad (31)$$

The gains for the velocity and the position control are tuned to 70 Hz and 10 Hz, respectively.

4.2.8. Load torque model

The contact between two sliding surfaces will generate random force fluctuations that are velocity dependent [32]. This effect is also present in the used ball bearings and can also be used in low-speed machine diagnostics [33]. A theoretical model for the acoustic emission between two sliding solids has been developed [32,34]. As the acoustic emissions are linked to system vibrations, this model can be used to describe the generated force fluctuations. According to this model, the amplitude of the fluctuations is proportional to

$$A = K_{AE} \sqrt{v} p^2, \quad (32)$$

where K_{AE} is a constant, v is the velocity of the sliding surfaces and p is the contact pressure on the surface of the asperities. Assuming a constant loading force of the ball bearings and a white frequency distribution [34], the torque fluctuation density can be modeled as

$$t_l(s) = K_N \sqrt{\omega}, \quad (33)$$

with the constant K_N , that needs to be determined by experiment, and the angular velocity ω .

5. Dynamic error budget

Before the system performance is analyzed, the model parameters are determined, the model is transformed in rotor coordinates, and the transfer functions of the disturbances are calculated.

5.1. System identification

A frequency response of the servo system is recorded for various velocities. This is realized by superimposing a chirp signal onto a constant ramp signal. The parameters of the used servo system are identified using the physical model of Section 4. A comparison between the measured frequency response and the simulated model response is shown in Fig. 3. In addition to the behavior predicted by the physical model, a resonance/anti-resonance at 90 Hz is fitted, which models a structural mode of the system. It is also visible in Fig. 3 that the system shows a very different behavior at different velocities, justifying the additional modeling effort. The parameters determined by the identification are given in Table 2.

5.2. Transfer functions for DEB

To simplify the system, all PSDs and transfer functions in the three-phase stator coordinate system (a, b, c) are transformed to the rotor coordinate system (d, q). The performance analysis of the system is conducted in two steps: First, all disturbance sources of the inner current control loop are referred to the mechanical rotor angle φ of the PMSM. Second, the cascaded motion control loop is added and the disturbance produced by the inner current control loop is weighted with its sensitivity function.

The inner current control loop of the torque producing q-component is shown in Fig. 4. The control loop of the d-component is identical, but has essentially no effect on the rotor position. Based on this representation and with the simplified notation $\frac{Y(s)}{X(s)} = G_{y,x}$, the sensitivity functions of the rotor position φ to the disturbances d are given by

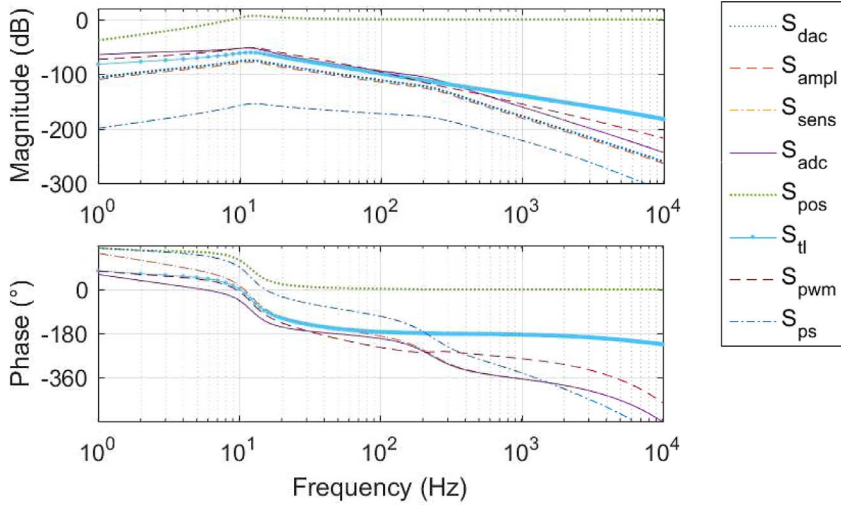


Fig. 5. Sensitivity functions of the rotor angle φ for the different disturbance acting in the position control system for 0.5°/s.

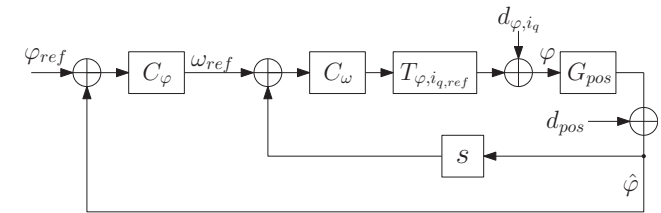


Fig. 6. Schematic of the cascaded FOC system with the complementary sensitivity function of the current control $T_{\varphi,i_q,ref}$ and the disturbance induced by the current control system d_{φ,i_q} .

5.3. Evaluation of the dynamic error budget

The noise performance of the considered servo system is simulated by using DEB [11,20]. For reasons of clarity, the results are plotted as cumulative power spectral densities (CPSDs) in logarithmic scale.

Fig. 7 shows the results for the linear operation mode of the amplifier for a velocity of (a) 0.01°/s and (b) 20°/s. For the low velocity case, the total position error n_{total} is dominated by the current sensor noise n_{sens} up to 300 Hz. In the higher frequency region, the noise of the encoder n_{pos} gets dominant. At higher velocity the influence of the ball bearing friction n_{ll} increases, resulting in a large increase of the position error.

Fig. 8 shows the results for the switched operation mode of the

amplifier for a velocity of (a) 0.01°/s and (b) 20°/s, with a switching frequency of 8 kHz. For the low velocity case, the position error is dominated by the current sensor noise, n_{sens} . However, due to the increased uncertainty of the current, this noise source dominates the spectrum up to 1 kHz. In the higher frequency region, the position noise from the encoder n_{pos} gets dominant. Also clearly visible is the influence of the PWM switching frequency, which leads to a sharp rise in the error. The increased noise of the amplifier n_{ampl} and the power supply n_{ps} are visible, but significantly smaller than the dominant noise sources. This appears to be in contrast to previously published work [13–15], identifying the power supply noise as a major noise source in switched amplifiers for audio applications. This difference can be explained by several factors: First, the applied amplifier is used with voltage feedback. This is beneficial by significantly increasing the PSRR in comparison to open loop switching amplifiers [14]. This can also be tested by considering a lower PSRR in the simulation. Doing so, the power supply noise starts to become a significant contributor to the overall noise, which is in agreement with the previously published results. Second, as shown in previous work [11], the amplifier noise in general has little influence on the position uncertainty of the investigated system, as the amplifier noise is suppressed by the current control loop. At higher velocities the influence of the ball bearings n_{ll} dominates all other noise sources.

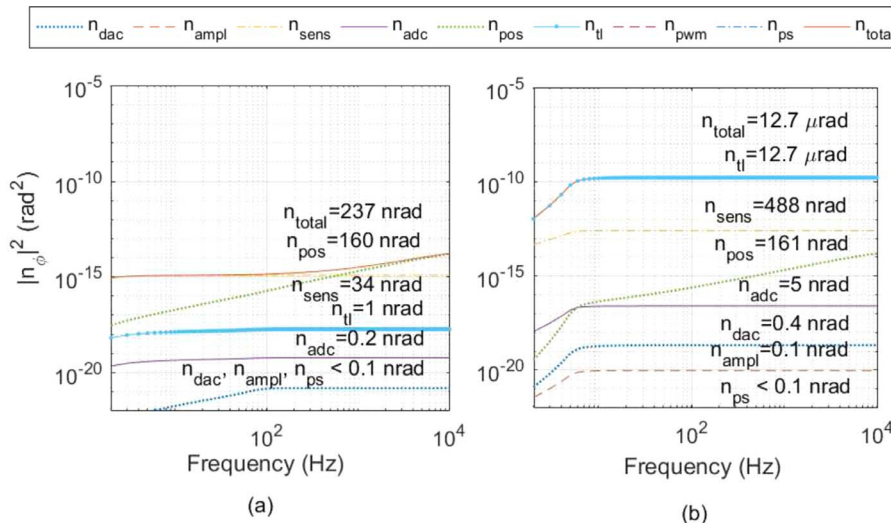


Fig. 7. Cumulative power spectral density of the position error for linear operation of the amplifier for a velocity of (a) 0.01°/s and (b) 20°/s.

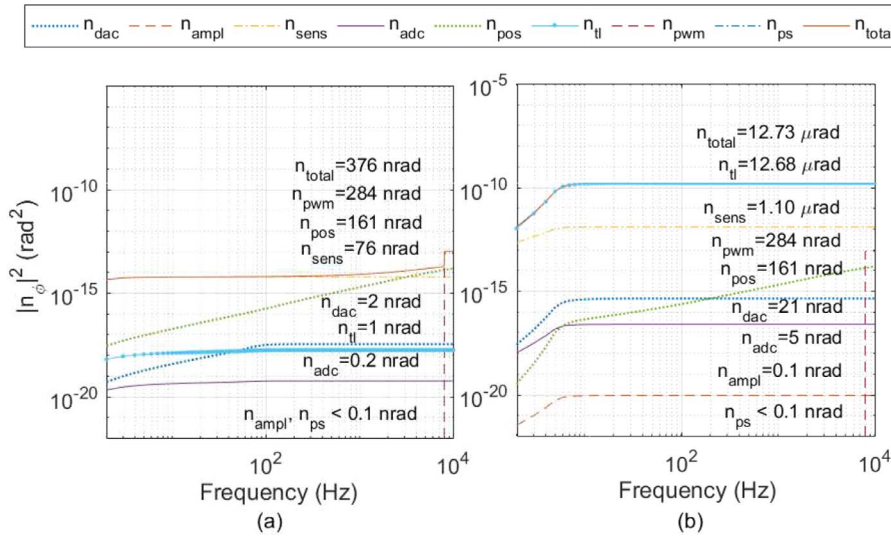


Fig. 8. Cumulative power spectral density of the position error for switched operation of the amplifier for a velocity of (a) 0.01°/s and (b) 20°/s and a switching frequency of 8 kHz.

6. Performance analysis and validation

To validate the simulation results, various performance criteria are analyzed and compared to experiments. First, the behavior of the system at different velocities is analyzed. The analysis is performed within a velocity range from 0.01°/s to 20°/s. A comparison of the measured data and the derived model is shown in Fig. 9. The measurement is performed in linear operation mode of the amplifier. The measurement data show a good match with the modeled behavior. In the lower velocity range, the distortions in the current loop are dominant. At higher velocity, the torque distortions due to the friction of the ball bearings start to become dominant, but only increase with the square root of the velocity.

To investigate the influence of the PWM frequency on the system performance, a variable PWM frequency generator is implemented on the FPGA of the control system. The PWM switching frequency is varied, and the position error at various velocities is recorded. The results are shown in Fig. 10. The position error is shown as a function of the velocity for various PWM switching frequencies, as well as for the linear operation mode. For clarity, the various measurements at different PWM frequencies are reduced to the mean value and its deviation, which are indicated by error bars.

It is visible in the measurement data as well as the modeled behavior that the variation between the linear and switched operation mode decreases as the velocity increases. This is mainly attributed to the fact that at higher velocities the torque disturbance due to the friction in the

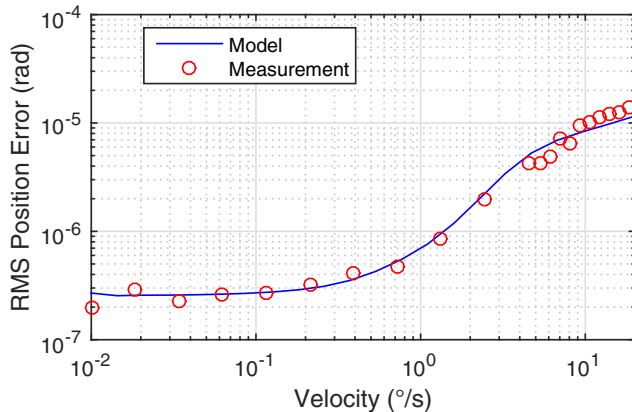


Fig. 9. RMS position error as function of the velocity simulated by DEB in comparison to measurement data for linear operation mode of the amplifier.

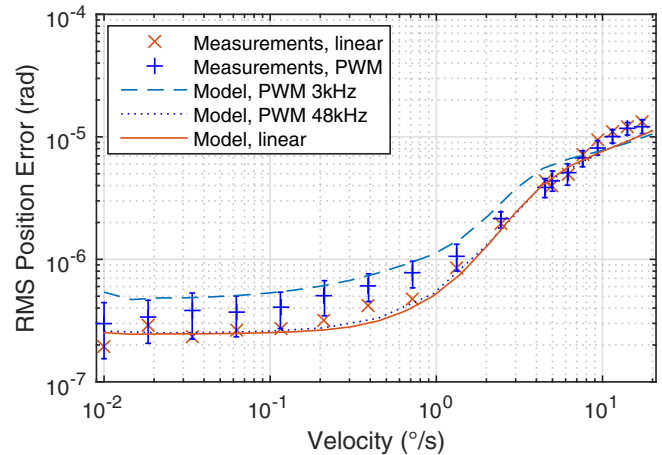


Fig. 10. RMS position error as function of the velocity in linear mode in comparison to switched mode. Error bars are shown for the measurements in switched mode, marking the measured range from 3 kHz to 48 kHz.

ball bearings is dominant. This is also confirmed by comparison with measurements of a similar system [35], where a comparable behavior is observed.

By considering the targeted system precision of 680 nrad for velocities up to 1°/s, Fig. 10 shows, that this requirement can be fulfilled in switched mode, with an appropriate switching frequency.

To identify an ideal switching frequency, the detailed results for a velocity of 1°/s are shown in Fig. 11. Also indicated in Fig. 11 is the corresponding position error for the linear mode and the targeted system precision of 680 nrad. Using the model, the minimal switching frequency that fulfills the targeted requirement is identified to be 15 kHz. A comparison with the measurement data as shown in Fig. 11 experimentally confirms this result. Using this relative low switching frequency implies maximal efficiency of the amplifier and also reduces the general speed requirements of the gate drivers and their corresponding control system. However, it should be noted that the audible vibrations, induced by the PWM, practically limit the switching frequency to values above 20 kHz. Comparing the efficiency at 20 kHz to the efficiency at 48 kHz, an improvement from 88 % to 92 % is achieved for a nominal load current of 0.5 A. Comparing the linear mode amplifier used in previous work [10,11], which achieves an efficiency of 5 % at nominal current, to the switched mode amplifier, the efficiency improvement is even more significant. It is also visible in Fig. 11 that the performance of the switched mode and the linear mode converge as

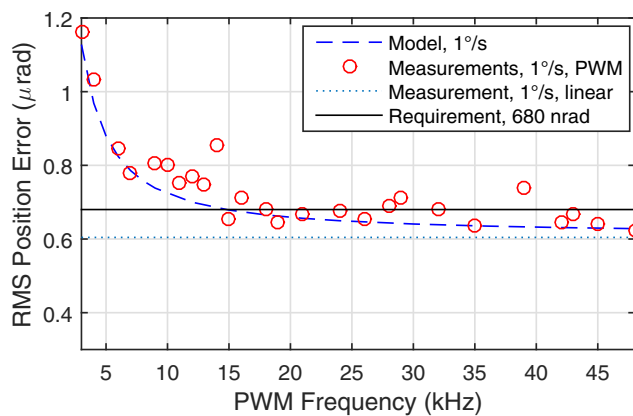


Fig. 11. RMS position error as function of the PWM frequency simulated by DEB in comparison to measurement data for a velocity of $1^\circ/\text{s}$.

the switching frequency increases. In contrast, for a switching frequency below 10 kHz, the error rapidly rises.

In summary, the performance analysis reveals a good match between the modeled behavior and the experimental data, confirming the used method and modeling approach. Applying DEB a minimal switching frequency of 15 kHz has been identified for the required precision of 680 nrad at $1^\circ/\text{s}$.

7. Conclusion

This paper presents the noise analysis of a pulse-width modulated permanent magnet synchronous motor of a direct drive rotational servo system. It is demonstrated that DEB can be applied to this complex, non-linear system in order to provide an accurate performance estimation. Physical modeling of the PMSM with linear as well as switched amplifier are used together with DEB to analyze the system. The analysis is performed to identify the minimal PWM switching frequency for achieving the required precision of 680 nrad at $1^\circ/\text{s}$. The minimal switching frequency is found to be 15 kHz, but for minimizing audible noise, a switching frequency of 20 kHz is chosen. Using this result, a system efficiency of 92 % is achieved, while maintaining the required system performance. The simulation results are compared to experimental data, confirming the applied method and modeling approach.

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